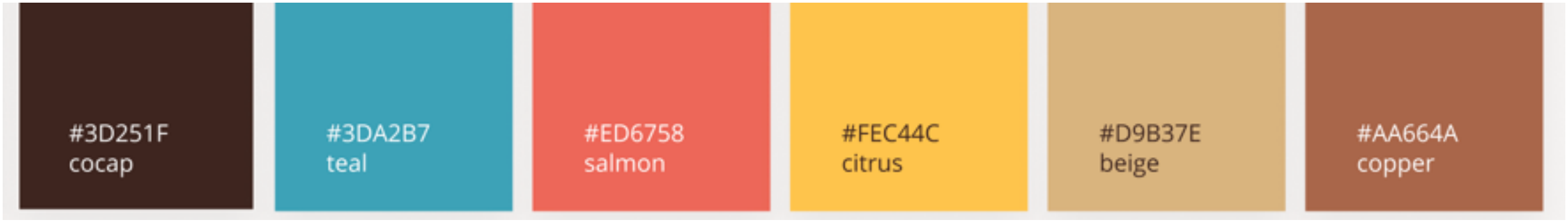


function

single 1-Lipschitz test

convex potentials



#3D251F
cocap

#3DA2B7
teal

#ED6758
salmon

#FEC44C
citrus

#D9B37E
beige

#AA664A
copper

Two Special Cases

Dual Problem

$$\sup_{(f,g) \in \mathcal{R}(c)} \int_X f(x) d\alpha(x) + \int_Y g(y) d\beta(y)$$

where $\mathcal{R}(c) = \{(f, g) \in \mathcal{C}(\mathcal{X}) \times \mathcal{C}(\mathcal{Y}) \mid \forall (x, y), f(x) + g(y) \leq c(x, y)\}$

1-Wasserstein (cost: $d(x, y)$)

$$W_1(\alpha, \beta) = \sup_{\|f\|_{\text{Lip}} \leq 1} \left(\int f d\alpha - \int f d\beta \right)$$

- Dual potentials collapse to a **single 1-Lipschitz test function** f .
- Intuition: f “probes” the difference between α and β
- Basis of Wasserstein GAN critic.

2-Wasserstein (cost: $d(x, y)^2$)

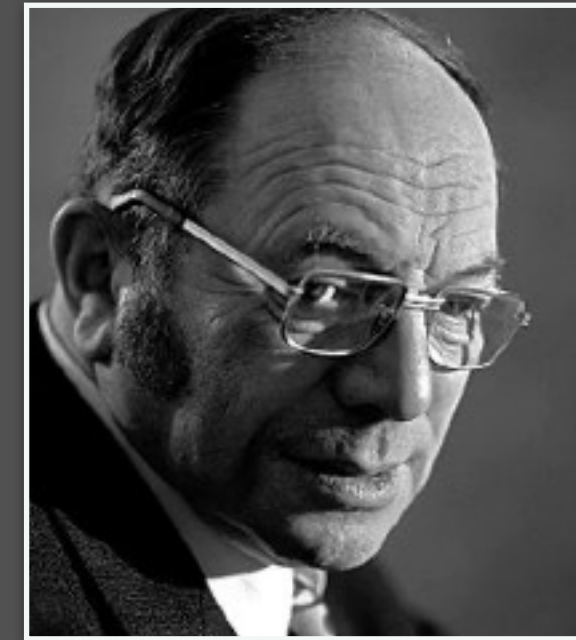
$$W_2^2(\alpha, \beta) = \int \|x\|^2 d\alpha(x) + \int \|y\|^2 d\beta(y) - 2 \inf_{\phi \text{ convex}} \left(\int \phi(x) d\alpha(x) + \int \phi^*(y) d\beta(y) \right)$$

- Dual involves **convex potentials** ϕ, ϕ^*
- Related to the Brenier map: optimal transport = gradient of a convex function.

Brenier's Theorem

Assumptions:

1. $c(x, y) = \|x - y\|^2$
2. α has a density $\rho_\alpha = \frac{d\alpha}{dx}$



Kantorovich's Problem:

(Continuous version)

$$\min_{\pi \in \Pi(\alpha, \beta)} \int_{X \times Y} c(x, y) d\pi(x, y)$$

Theorem [Brenier, 1991]:

1. There exists a **unique** coupling π that solves the Kantorovich problem
2. The coupling is deterministic, supported on the graph of a Monge map: $\pi = (\text{Id}, T)_\# \alpha$
3. T is the gradient of a convex function

Formally: the unique $T = \nabla \varphi$ such that φ is **convex** and $(\nabla \varphi)_\# \alpha = \beta$

